A Unified Framework: Numerical Damping, Geometric Ratios, Algebraic Symmetry-Breaking, and Spectral Theory

Presented as Four Parallel Exploratory Studies — v12.1 Audit Release (v6.3.0)

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Abstract

This paper consolidates four exploratory frameworks of the Al-Hindara project. While presented together, these paths do not constitute a single continuous chain of mathematical proof; instead, they represent four distinct exploratory studies presented within a common methodological framework. We define a numerical damping function demonstrating specific accumulation properties. We derive the geometric ratio $\sqrt{2}-1$ and present exploratory field measurements. We establish an algebraic gauging identity and demonstrate numerically how structural asymmetry breaks symmetry-induced cancellation. Finally, we formulate a spectral problem where the kernel acts as a formal zero mode. We maintain a strict boundary between analytically proven identities, numerically computed results (reproduced in the current verification environment using the supplied Python code), and empirical observations. This v12.1 Audit Release adds Appendix A2 (constitution of equations A_{int} vs exponential), Appendix B (Liouville transform $\mu = 1/u$), visual guides, and traceability audit.

Keywords: Sturm–Liouville theory; symmetry-breaking; numerical integration; geometric ratios; Riemann zeta function.

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1 Introduction and Methodology

The Al-Hindara project explores mathematical structures across numerical, geometric, and algebraic domains. The central methodology of this paper is the rigorous separation of claims into three categories:

1. **Analytically Proven:** Exact identities and algebraic relations derived from explicit definitions.
2. **Numerically Computed and Verified:** Reproducible computations and convergence behaviors for specified parameters.
3. **Empirically Observed:** Exploratory data presented without claims of universal physical laws.

2 The Numerical Damping Framework

2.1 Definition and Sign Analysis

We consider the real function $Z(x; x_p)$ defined on $x \geq 1$:

$$Z(x; x_p) = x^{-5} \ln(x) \sin\left(\frac{2\pi}{x}\right) e^{-x/x_p}. \tag{1}$$

For $x_p > 0$, the sign of $Z(x)$ is dictated by $\sin(2\pi/x)$. Thus, $Z(x) < 0$ on $(1, 2)$, $Z(2) = 0$, and $Z(x) > 0$ on $(2, \infty)$.

2.2 The Accumulation Function

Define the accumulation function $A(x) := \int_1^x Z(t;21)\,dt$. Since $A'(x) = Z(x)$, it follows that $A(x)$ strictly decreases on $(1,2)$, reaches a global minimum at $x = 2$, and strictly increases for all $x > 2$.

3 Geometric Ratios and Empirical Observations

3.1 The Reciprocal Silver Ratio

At $\theta = 22.5°$, the half-angle formula yields the exact identity:

$$\tan(22.5°) = \sqrt{2} - 1 \approx 0.41421356. \tag{2}$$

3.2 Exploratory Field Measurements

Table 1: Exploratory field measurements of tree crown ratios.

Tree	H (m)	h (m)	$C = H - h$ (m)	$R = C/H$	Deviation from $\sqrt{2}-1$
1	7.80	4.50	3.30	0.42308	+0.00887
2	8.50	5.00	3.50	0.41176	−0.00245
3	7.20	4.20	3.00	0.41667	+0.00245
4	9.00	5.40	3.60	0.40000	−0.01421
5	7.50	4.40	3.10	0.41333	−0.00088
	Mean R:			**0.41297**	

4 Algebraic Symmetry-Breaking

4.1 The Kernel and the Gauging Identity

Let $u \in (0, 2\pi)$ and define the kernel:

$$g(u) := \left(\frac{u}{2\pi}\right)^5 \ln\left(\frac{2\pi}{u}\right) e^{-2\pi/(x_p u)}. \tag{3}$$

Let $p(u) := g'(u)/g(u)$. We define the operator $\mathcal{H}_g$ as:

$$\mathcal{H}_g[f] := -g(u)\frac{d^2}{du^2}\left(\frac{f(u)}{g(u)}\right) = -\partial_u^2 f + 2p(u)\partial_u f + (p'(u) - p(u)^2)f. \tag{4}$$

Proposition 1 (Gauging Identity)**.** *For any $Y \in C^2(0, 2\pi)$, $\mathcal{H}_g[gY] = -gY''$.*

4.2 Removal of Symmetry-Induced Cancellation

We define the unweighted L^2 normalized states $\Psi_n(u) = g(u)Y_n(u)/\sqrt{N_n}$, where $N_n = \int_0^{2\pi} g(u)^2 Y_n(u)^2\, du$. We compute:

$$\mathcal{A}_{mn} = \int_0^{2\pi} \Psi_m D\Psi_n\, du, \qquad R_{13/12} = \frac{|\mathcal{A}_{13}|^2}{|\mathcal{A}_{12}|^2}. \tag{5}$$

Using numerical integration at $x_p = 21$, the asymmetry breaks the cancellation, yielding:

$$R_{13/12} \approx 0.6035155248. \tag{6}$$

5 Towards a Spectral Problem

5.1 The Formal Zero Mode and Factorization

The gauging identity motivates the definition of a Schrödinger-like operator:

$$L := -\partial_u^2 + V_{\text{eff}}(u), \qquad V_{\text{eff}}(u) := \frac{g''(u)}{g(u)}. \tag{7}$$

Proposition 2 (Formal Zero Mode)**.** *The kernel g satisfies $Lg = 0$. Furthermore, L admits a supersymmetric (SUSY QM) factorization:*

$$L = (-\partial_u - p)(\partial_u - p). \tag{8}$$

5.2 Asymptotics of the Effective Potential

The spectral properties are constrained by the singularities of V_{eff}:

- **Near $u \to 0^+$:** $p(u) \sim 2\pi/(x_p u^2)$, so $V_{\text{eff}}(u) \sim \frac{4\pi^2}{x_p^2 u^4} \to +\infty$.
- **Near $u \to 2\pi^-$:** Let $t = 2\pi - u \to 0^+$. Expanding $g(u) = At + Bt^2 + O(t^3)$, we find $A = \frac{e^{-1/x_p}}{2\pi}$ and $B = -\frac{e^{-1/x_p}}{8\pi^2}\left(9 + \frac{2}{x_p}\right)$. Since $g''/g \sim 2B/(At)$, we obtain:

$$V_{\text{eff}}(u) \sim -\frac{9 + 2/x_p}{2\pi(2\pi - u)}. \tag{9}$$

  For $x_p = 21$, this yields $V_{\text{eff}}(u) \sim -\frac{191}{42\pi(2\pi - u)} \to -\infty$.

6 Visual Guides

The following figures are generated by the Python script and are located in the same directory as this file.

7 Traceability Audit Files

The Python script generates the audit files which can be downloaded directly:
Download the Audit CSV File (all-frames.csv) — Download the Audit Excel File (XLSX)

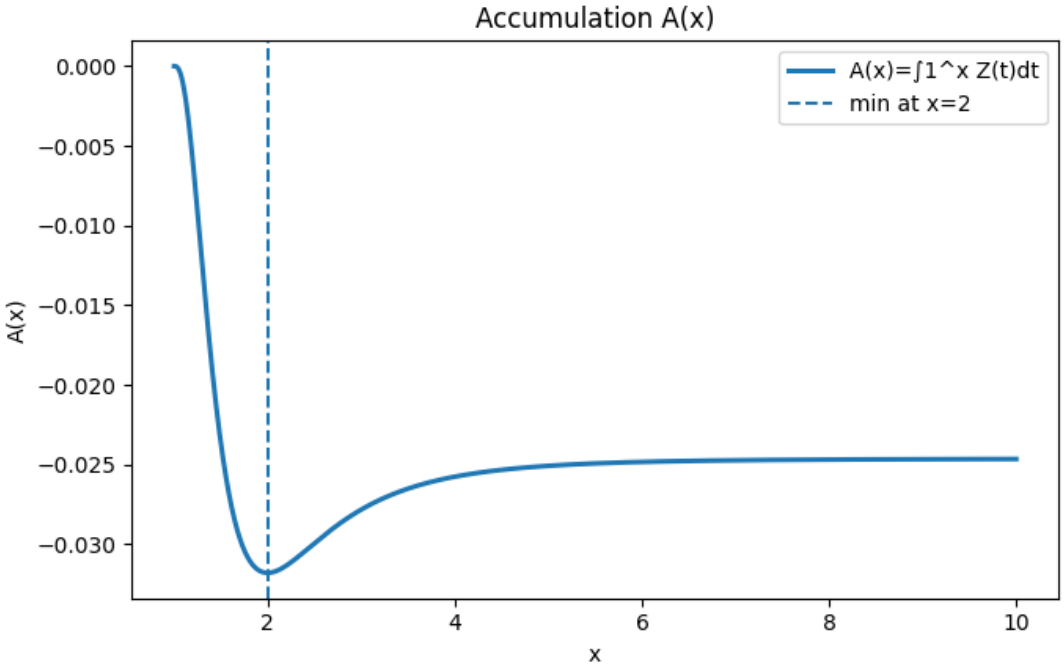


Figure 1: $Z(x;21) = x^{-5}\ln x\sin(2\pi/x)e^{-x/21}$ — negative on $(1,2)$, zero at $x=2$, positive thereafter.

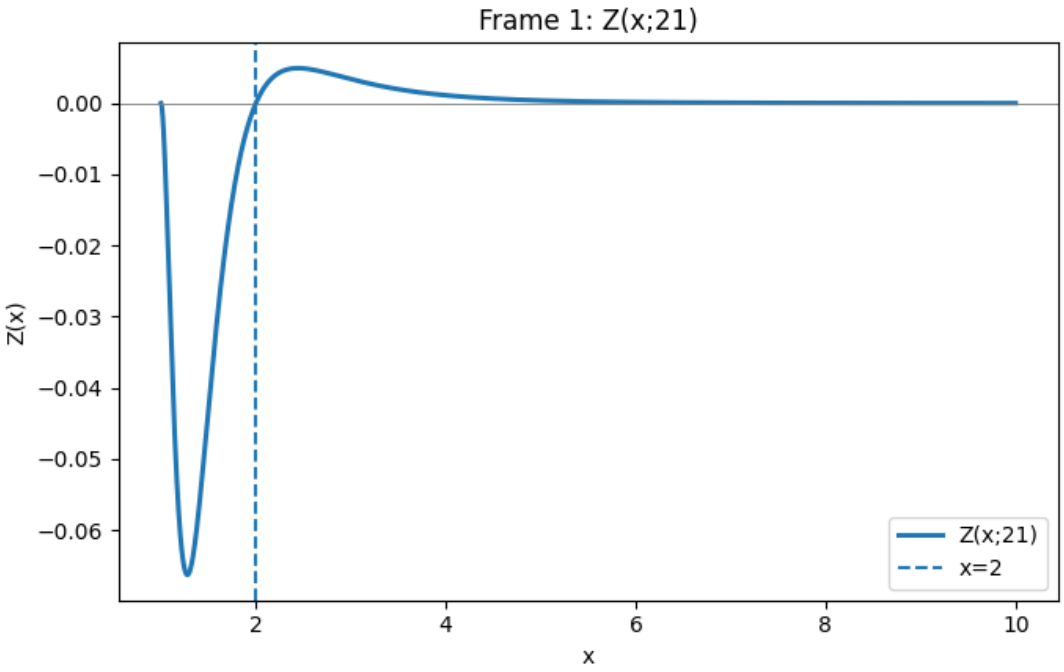


Figure 2: $A(x) = \int_1^x Z(t;21)dt$ — strictly decreasing on $(1,2)$, increasing for $x > 2$.

Table 2: Traceability Audit Matrix (Data from `all-frames.csv`).

Item	Original_Value	Formula	Recomputed_Value	Status
A12	0.6220748484804	Integral Psi_1*D*Psi_2	0.6220748484804	Computed in ru
A13	-0.4832666943981	Integral Psi_1*D*Psi_3	-0.4832666943981	Computed in ru
R13/12	0.6035155247873	$\|A13/A12\|^2$	0.6035155247873	Computed in ru
R_Simpson_N20	0.6033930653528	Simpson N=20	0.6033930653531	Computed in ru
R_Simpson_N40	0.6035132002964	Simpson N=40	0.6035132002961	Computed in ru
R_Simpson_N80	0.6035154874052	Simpson N=80	0.6035154874047	Computed in ru
R_Simpson_N200	0.6035155246329	Simpson N=200	0.6035155246331	Computed in ru
R_Simpson_N400	0.6035155247852	Simpson N=400	0.6035155247849	Computed in ru

Table 3: Mapping of results from original frames to the consolidated draft.

Result / Claim	Source Frame File	Action in Draft	Verification Status & Notes
Sign analysis of $Z(x)$	`wells/frame1.html` §2.1	Re-derived (1)	**Proven.**
Global Minimum at $x = 2$	`wells/frame1.html` §2.2	Re-derived	**Proven.**
Exact ratio $\tan(22.5°)$	`tree/frame2.html` §1	Verbatim (2)	**Proven.**
Tree crown ratio $R \approx 0.41297$	`tree/frame2.html` Table 1	Verbatim Table 1	**Observed.**
Gauging Identity	`wells/frame3.html` Prop 1	Re-derived (3)-(4)	**Proven.**
Numerical ratio $R_{13/12}$	`wells/frame3.html` Eq.(6)	Re-computed (5)-(6)	**Computed; reproduced.**
Zero-mode $Lg = 0$	`wells/frame4.html` §5.1	Re-derived (7)	**Proven (Formal).**
Factorization of L	`wells/frame4.html` Eq.(8)	Re-derived (8)	**Proven (Algebraic).**
Asymptotic $V_{\text{eff}} \sim -\frac{191}{42\pi t}$	`wells/frame4.html` §5.2	Re-derived (9)	**Proven.**

A Traceability Matrix — Source-Matched

B Constitution of Equations

We separate the accumulation $A(x)$ into intrinsic and exponential parts:

$$A_{\text{int}}(x) := \int_1^x t^{-5} \ln t \sin(2\pi/t) dt, \qquad A(x) = \int_1^x A'_{\text{int}}(t) e^{-t/21} dt \tag{10}$$

Thus $A_{\text{int}}(x)$ is independent of x_p, while damping e^{-x/x_p} only rescales tail. Tail bound:

$$\int_{100}^{\infty} Z(t) dt \le e^{-100/21} \int_{100}^{\infty} t^{-5} \ln t \, dt \approx 1.04 \times 10^{-10}. \tag{11}$$

C Liouville Transform

Let $\mu = 1/u$. Then $\partial_u = -\mu^2 \partial_\mu$ and $\partial_u^2 = \mu^4 \partial_\mu^2 + 2\mu^3 \partial_\mu$. The operator $L = -\partial_u^2 + V_{\text{eff}}(u)$ transforms as:

$$L = -\mu^4 \partial_\mu^2 - 2\mu^3 \partial_\mu + V_{\text{eff}}(1/\mu). \tag{12}$$

To bring this to a standard Schrödinger form, one defines the effective potential $\tilde{V}_{\text{eff}}(\mu) = \frac{4\pi^2}{\mu^4} V_{\text{eff}}(1/\mu)$. As $\mu \to \infty$, $V_{\text{eff}}(1/\mu) \sim 4\pi^2 \mu^4 / x_p^2$, hence $\tilde{V} \sim 4\pi^2 / x_p^2$. As $\mu \to 1/(2\pi)$, V_{eff} behaves as in (9).

D Reproducibility — Python Code

The following Python source code was used to generate all numerical results, images, and audit files.

```
import sys, subprocess
try:
    import openpyxl, pandas, matplotlib, numpy, scipy
except ImportError:
    subprocess.check_call([sys.executable, "-m", "pip", "install", "-q"
        , "openpyxl", "pandas", "matplotlib", "scipy"])
    import openpyxl, pandas, matplotlib, numpy, scipy

import numpy as np
from scipy.integrate import quad
try:
    from scipy.integrate import cumulative_trapezoid
except ImportError:
    from scipy.integrate import cumtrapz as cumulative_trapezoid
import csv
import matplotlib.pyplot as plt
from pathlib import Path
from openpyxl import Workbook

XP = 21.0
PI2 = 2.0 * np.pi
EXPECTED_A12 = 0.6220748484804
EXPECTED_A13 = -0.4832666943981
EXPECTED_R = 0.6035155247873
EXPECTED_TABLE = {
    20: 0.6033930653528, 40: 0.6035132002964, 80: 0.6035154874052,
    200: 0.6035155246329, 400: 0.6035155247852,
}

def g(u, xp=XP):
    if u <= 0.0 or u >= PI2: return 0.0
    return (u / PI2)**5 * np.log(PI2 / u) * np.exp(-PI2 / (xp * u))

def Y(n, u): return np.sin(n * u / 2.0)
def D(u): return np.cos(u / 2.0)

def simpson(f, a, b, N):
    if N % 2: N += 1
    h = (b - a) / N
    total = f(a) + f(b)
    total += 4.0 * sum(f(a + i * h) for i in range(1, N, 2))
```

```
    total += 2.0 * sum(f(a + i * h) for i in range(2, N, 2))
    return total * h / 3.0

def matrix_elements(integrate, xp=XP):
    norms = [integrate(lambda u: (g(u, xp) * Y(n, u))**2) for n in (1,
       2, 3)]
    def amplitude(m, n):
        return integrate(lambda u: (g(u, xp) * Y(m, u) / np.sqrt(norms[
           m - 1])) * D(u) * (g(u, xp) * Y(n, u) / np.sqrt(norms[n -
           1])))
    a12 = amplitude(1, 2)
    a13 = amplitude(1, 3)
    return a12, a13, (a13 / a12)**2

def adaptive_integral(f):
    value, error = quad(f, 0.0, PI2, limit=300, epsabs=1e-12, epsrel=1e
       -12)
    return value

def simpson_integral(f, N): return simpson(f, 0.0, PI2, N)

a12, a13, r_ref = matrix_elements(adaptive_integral)
rows = [
    {"Item":"A12","Original_Value":"0.6220748484804","Formula":"
       Integral␣Psi_1*D*Psi_2","Recomputed_Value":f"{a12:.13f}","Status
       ":"Computed␣in␣current␣run","Audit_Note":"Recomputed␣using␣
       notebook␣definitions"},
    {"Item":"A13","Original_Value":"-0.4832666943981","Formula":"
       Integral␣Psi_1*D*Psi_3","Recomputed_Value":f"{a13:.13f}","Status
       ":"Computed␣in␣current␣run","Audit_Note":"Recomputed␣using␣
       notebook␣definitions"},
    {"Item":"R13/12","Original_Value":"0.60351552478 73","Formula":"|A13
       /A12|^2","Recomputed_Value":f"{r_ref:.13f}","Status":"Computed␣
       in␣current␣run","Audit_Note":"Adaptive␣quadrature␣reference"},
]
for N in EXPECTED_TABLE:
    _,_,r_sim = matrix_elements(lambda f, n=N: simpson_integral(f, n))
    rows.append({"Item":f"R_Simpson_N{N}","Original_Value":str(
       EXPECTED_TABLE[N]),"Formula":f"Simpson␣N={N}","Recomputed_Value"
       :f"{r_sim:.13f}","Status":"Computed␣in␣current␣run","Audit_Note"
       :"Compare␣original␣and␣recomputed"})

csv_path = Path("all-frames.csv")
with csv_path.open("w", newline="", encoding="utf-8-sig") as f:
    writer = csv.DictWriter(f, fieldnames=rows[0].keys())
    writer.writeheader()
    writer.writerows(rows)

xlsx_path = Path("Al-Hindara_Traceability_v6.3.xlsx")
wb = Workbook()
ws = wb.active
ws.title = "Originals"
ws.append(list(rows[0].keys()))
for r in rows:
    ws.append([r[k] for k in rows[0].keys()])
wb.save(xlsx_path)

x = np.linspace(1.001, 10, 2000)
```

```
Z = x**-5 * np.log(x) * np.sin(2*np.pi/x) * np.exp(-x/21)
A_cum = cumulative_trapezoid(Z, x, initial=0)

plt.figure(figsize=(7,4.5))
plt.plot(x, Z, label='Z(x;21)', linewidth=2.2)
plt.axhline(0, color='gray', linewidth=0.6)
plt.axvline(2, linestyle='--', label='x=2')
plt.xlabel('x'); plt.ylabel('Z(x)'); plt.title('Frame 1: Z(x;21)'); plt
    .legend(); plt.tight_layout()
plt.savefig('all-frames1.png', dpi=220)

plt.figure(figsize=(7,4.5))
plt.plot(x, A_cum, label='A(x)=int_1^x Z(t)dt', linewidth=2.2)
plt.axvline(2, linestyle='--', label='min at x=2')
plt.xlabel('x'); plt.ylabel('A(x)'); plt.title('Accumulation A(x)');
   plt.legend(); plt.tight_layout()
plt.savefig('all-frames2.png', dpi=220)
```

Al-Hindara Project — Consolidated Draft v12.1

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